

ENVS-193DS Final

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[Link to repo](#)

```
# read in packages
library(tidyverse)
library(here)
library(DHARMA)

# reading in data and creating new object
nest <- read_csv(
  here("data", "nest_data_final.csv"))
```

Problem 1. Research writing

Problem 2. Data Analysis

a.

```
nests_clean <- nest |>
  select(ant_species, height_cm, case_control) |>
  filter(ant_species %in% c("RRB", "BBR", "AB")) |>
```

```
mutate(sci_name = case_when(
  ant_species == "RRB" ~ "Crematogaster sjostedti",
  ant_species == "BBR" ~ "Crematogaster mimosae",
  ant_species == "AB" ~ "Crematogaster nigriceps"),
  ant_species = factor(ant_species,
    levels = c("RRB", "BBR", "AB"))
)

str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ ant_species : Factor w/ 3 levels "RRB","BBR","AB": 1 1 1 1 3 1 2 1 3 2 ...
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ sci_name    : chr [1:270] "Crematogaster sjostedti" "Crematogaster sjostedti" "Crematogaster sjostedti" ...
```

```
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
  ant_species height_cm case_control sci_name
  <fct>         <dbl>         <dbl> <chr>
1 RRB           194.             0 Crematogaster sjostedti
2 RRB           247             1 Crematogaster sjostedti
3 RRB           134.             0 Crematogaster sjostedti
4 BBR           170             0 Crematogaster mimosae
5 RRB           208             0 Crematogaster sjostedti
6 RRB           288.             1 Crematogaster sjostedti
7 AB            153             0 Crematogaster nigriceps
8 RRB           598             0 Crematogaster sjostedti
9 RRB           129             0 Crematogaster sjostedti
10 BBR           71             0 Crematogaster mimosae
```